

Sleep Scores: Your Smartwatch Is Not a Sleep Laboratory (And That's Fine)

The sleep score should answer “how restorative was last night’s sleep?”, not “am I ready to train?”. In practice this means sleep duration, continuity, regularity, and a small bit of physiology. The important point from the literature is that sleep trackers are usually more reliable for broad sleep-wake behaviour and trends than for exact metrics.

Wrist actigraphy (method estimating sleep/wake using movement data from a wrist device) is strong for detecting sleep overall, but weaker for detecting wake, and consumer devices infer stages from movement and cardiorespiratory signals rather than measuring brain activity directly (obviously). [1]

We’ll start by understanding what the mainstream providers currently do to derive their sleep scores. They all follow a common pattern: mixing sleep amount, sleep quality/continuity, and some form of physiological recovery signal, but they differ in how much they expose the logic.

Fitbit’s sleep score has typically been described through dimensions such as duration, restoration, and restlessness, and Google’s more recent redesign appears to be moving toward a more transparent version that shows total sleep, time to reach “sound sleep,” interruptions, and full awakenings, in some cases with comparisons by age and gender.

This is a sensible evolution because users can understand it (one of the bits of the health app I don’t hate). [2]

Apple entered this space later with a Sleep Score in watchOS 26. Public reports say the score uses signals such as heart rate, wrist temperature, blood oxygen, and respiratory data, while other reporting indicates Apple also weights more actionable behavioural features like bedtime consistency and time in bed.

The general direction is relatively clear: the company is treating sleep as both a physiological and a behavioural system. [3]

Garmin’s sleep ecosystem is less centered on one single number and more on the surrounding tools. Public guides emphasise overnight stress, Body Battery, respiration rate, and their Sleep Coach, which adjusts sleep goals based on activity, HRV, and naps.

This serves as a useful reminder that a good sleep product is not just a number in the morning; it is also guidance on how much sleep the person likely needs, we want this number to help the user improve their readiness. [4]

WHOOP leans even harder into the “sleep as recovery fuel” framing. WHOOP directly links higher strain to higher Sleep Need, and the company’s Healthspan work treats sleep consistency and sleep duration as core long-term health metrics. [5]

Moving to the scientific literature, we see 3 main conclusions.

Firstly, sleep quantity and regularity are more robust than fine-grained sleep staging in wearables. Wrist actigraphy is good at identifying sleep versus wake overall, but it has difficulty with

wake detection, and stage-level inference remains much less secure.

More advanced wearable staging models using PPG and contextual windows can achieve good performance, but are still not the same as lab PSG (polysomnography - proper lab sleep studies). That doesn't make them useless but it means we should weight their outputs conservatively. [6]

The second is that sleep regularity matters far more than many apps used to admit. Large observational work summarised in recent reports found that irregular sleep schedules were strongly associated with worse cardiovascular outcomes, and other large-scale work linked higher sleep regularity to a lower mortality risk.

From a product-design point of view, this means regularity should not be a tiny bonus metric but rather a major component of the score. [7]

The third is that wearables are most useful when they show trends and behaviour links, not when they pretend to be medical sleep labs. Sleep specialists quoted in recent reports make this point repeatedly: these devices can help people identify patterns around alcohol, late eating, bedtime drift, illness, and disturbance, but exact stage numbers should not be treated as truth. [8]

The more important question for us is not just "are wearables imperfect?", because they obviously are. The more useful question is which parts of sleep are reliable enough and important enough to drive a 'sleep score'. This is where I think the feature order becomes much clearer.

For newer products such as the Fitbit Air, we should be more optimistic than a generic "wrist trackers are bad" argument. Scientific evaluation videos compare the device against stronger reference measurements and suggest that these devices perform well enough to be useful, even if not perfect.

The important distinction is that this should increase our confidence in wearables as a practical data source, but it should not change the basic scientific hierarchy. Even a good wearable is still more trustworthy for sleep duration, timing, continuity and broad sleep behaviour than it is for exact REM/deep/light stage breakdowns. [9]

The strongest order of importance looks like this:

- Sleep duration: This should likely be the largest part of the score. If the user does not get enough total sleep, the rest of the score should not be able to fully rescue it. Sleep stages can look "good" within a short sleep, but 4 hours of sleep is still, at the end of the day, 4 hours of sleep.
- Sleep regularity: This is the most underweighted feature in many current apps. Large cohort work has found sleep regularity to be strongly associated with mortality and cardiovascular outcomes, in some cases more strongly than just duration alone. From a scoring perspective, this means consistency should not be a tiny bonus. It should be a major block of the score. [10] [11]
- Continuity and awakenings: Sleep that is constantly interrupted is not the same as one continuous block of sleep. This is where wake after sleep onset, number of awakenings, restlessness, and sleep efficiency matter. Wearables are not perfect at wake detection, but this is still more reliable and actionable than obsessing over the exact stage percentages.
- Timing and sleep onset: Bedtime drift, very late sleep, and taking a long time to fall asleep should also matter, but slightly less than duration, regularity and continuity. Timing is important because it reflects circadian stability, not just "when did I happen to go to bed".
- Overnight physiology: HRV, RHR, respiratory rate, SpO₂ and temperature can help explain why sleep may have been poor or why the body looked stressed overnight. However,

these should be smaller components of the sleep score than they are in a readiness score. They are more useful as context than as the definition of the sleep's quality.

- Sleep stages: REM and deep sleep should be included, but carefully. There is evidence that longer deep sleep, higher sleep efficiency and earlier REM latency are associated with more favourable wearable-derived health markers, but stage detection is still less secure than duration and timing. The key point is: sleep stages are useful supporting evidence, not the backbone of our score. [12] [13]

Given this, the sleep score should weight duration at 35%, regularity at 25%, continuity at 20%, timing and sleep onset at 10%, overnight physiology at 5%, and sleep stages at 5%.

These weights should be fixed for the initial algorithm. Future validation can test whether stage weighting should vary by provider, but stages should always carry less weight than duration, regularity and continuity.

The sleep score should be partly personalised, although not in the same way as readiness or strain. Duration should use the personalised core sleep need defined in Sleep Need, Debt, Recovery and Stages. Regularity should be strongly personalised because the important question is whether the user is consistent relative to their own life. Continuity should also be compared with the person's baseline because some people naturally wake more often than others.

The score should also have a confidence layer. If duration and timing are available but stages are missing, the software should still calculate a sleep score as the strongest features are still present. If sleep stages look strange but duration, regularity and continuity are good, the score shouldn't collapse.

If duration is short, bedtime is late, and awakenings are high, the score should drop even if the device claims a decent amount of REM. This is the core principle: the sleep score should reward the most reliable and most important sleep features first.

This gives us a sleep score that answers the right question: "how restorative was last night's sleep?", rather than pretending the wearable is a medical sleep lab or accidentally turning the sleep score into a readiness score.

These principles describe what the sleep score should value. To turn them into an implementable algorithm, each component also needs a defined 0-100 scale, a calculation method and clear rules for missing or unreliable data.

Across the components, a score of 100 means the observed night gave no meaningful reason to reduce that component. It does not mean biologically perfect sleep. A score around 80 represents broadly adequate sleep with some room for improvement, around 60 indicates a meaningful problem, and scores below 40 should be reserved for a large shortfall or severe disruption. Straight-line interpolation should be used between the anchor values described below.

Calculating sleep duration

Sleep duration accounts for 35% of the score. Sleep achieved comes from the valid 24-hour sleep day defined in the accompanying sleep paper, while the comparison target is core sleep need rather than the larger recovery opportunity. Recovery opportunity is advice about how much time to allow for sleep; failing to complete all of it should not manufacture a new debt on top of the existing one.

The duration calculation should use the shortfall remaining after the 30-minute allowance described in that paper:

$$\text{duration shortfall} = \max(0, \text{core sleep need} - \text{valid sleep achieved} - 30 \text{ minutes})$$

The duration component should then use these anchor points, with straight-line interpolation between them:

- 0 minutes of scored shortfall: 100.
- 30 minutes: 95.
- 60 minutes: 90.
- 90 minutes: 80.
- 120 minutes: 65.
- 180 minutes: 35.
- 240 minutes: 10.
- 300 minutes or more: 0.

This makes a small miss visible without treating it like a disaster, while a night that is 3 or 4 hours short cannot hide behind good-looking continuity or stage data. Sleep above core need should not lose points in this component.

A night more than 2 hours above need can be marked as unusually long and explained alongside illness or recent restriction, but long sleep is not automatically bad sleep. The companion paper already explains why extra sleep can't create a positive debt balance for later.

Calculating sleep regularity

Sleep regularity should be understood as a short rolling behaviour rather than a component contained entirely inside one night. The Personal Metric Baselines paper explains why bedtime and wake time need circular calculations, separate workday and free-day references, and a stable device source.

Where a full sleep-wake timeline is available, the Sleep Regularity Index (SRI) is useful for the longer view because it asks how often the person is in the same sleep or wake state at points 24 hours apart. It is sensitive to naps, awakenings and all-nighters, while a simple standard deviation of bedtime is not.

Composite Phase Deviation is useful when the focus is how much the latest sleep midpoint differs from both the previous day and the person's usual timing. These measures answer slightly different questions, so we should not casually average every regularity metric we can calculate. [15]

These measures are useful for longer-term reporting, although the nightly component needs a simpler and more interpretable calculation. It should use the actual bedtime and wake-time drift from the correct day-type reference. Calculate their circular differences, then take the mean:

$$\text{schedule drift} = \frac{|\text{bedtime drift}| + |\text{wake-time drift}|}{2}$$

The regularity score should use these anchors:

- 0 to 30 minutes of schedule drift: 100.
- 60 minutes: 90.
- 90 minutes: 75.
- 120 minutes: 55.
- 180 minutes: 20.
- 240 minutes or more: 0.

Straight-line interpolation fills the gaps. At least 7 valid comparable nights are needed before this personal component can be treated as established, with 28 preferred. Before then, the latest bedtime and wake time can still be shown, but the regularity component is provisional.

The 7-day SRI should be reported beside the component as context instead of being used as a second penalty for the same behaviour. An SRI of 100 means perfectly repeated sleep-wake timing, although cohort cutoffs should not be treated as universal nightly score boundaries. [21]

Calculating continuity

Continuity should focus on what happened after the sleep session actually began. This keeps it separate from sleep onset latency. The main measurement is wake after sleep onset (WASO), supported by the number of awakenings lasting more than 5 minutes.

Sleep efficiency should be shown, but it should not be averaged as another independent input because efficiency, WASO and awakenings are largely different descriptions of the same lost sleep.

Expert consensus judged WASO of 20 minutes or less as a sign of good sleep quality across the lifespan. For most adult groups, 41 to 50 minutes was already considered poor, with 51 minutes or more clearly outside good sleep quality.

Sleep efficiency of at least 85% was considered good, while 74% or lower was poor for most age groups. The same review found 0 or 1 awakening lasting more than 5 minutes was good, while 4 or more was poor; 2 awakenings remained acceptable for older adults. [14]

WASO can therefore drive the continuity score using these anchors:

- 0 to 20 minutes: 100.
- 30 minutes: 90.
- 40 minutes: 75.
- 50 minutes: 55.
- 60 minutes: 35.
- 90 minutes or more: 0.

The score is then capped when the supporting measurements show a clearly fragmented night. Maintenance efficiency below 85% caps continuity at 80, below 75% caps it at 40, and below 65% caps it at 20. Four or more awakenings lasting over 5 minutes cap it at 60.

2 or 3 such awakenings cap it at 85, except that 2 awakenings produce no cap for somebody aged 65+. These are caps rather than separately weighted penalties, which stops the same wake time being counted 3 times.

Wearables are often weakest at detecting quiet wake, so provider-specific personal comparison still matters. A night that is more than 2 spreads worse than the person's 28-night WASO or maintenance-efficiency reference should be flagged even if it narrowly remains inside a population anchor.

The population thresholds give the curve a sensible shape; the personal reference tells us when the night was unusual for this user. Normal polysomnography also changes with age: across adult decades, sleep efficiency falls and WASO rises, so older sleep should not be judged as though it came from a healthy 20-year-old. [16]

Calculating timing and sleep onset

Timing and sleep onset account for 10% of the score. This component has 2 possible inputs. Sleep latency measures the time from trying to sleep to the first sustained sleep.

Circadian alignment measures how far the sleep midpoint sits from the person's chosen or learned target window. Regularity asks whether the schedule repeats; alignment asks whether the schedule happened near the window that normally fits the person's life and daytime function.

The latency subscore should use the consensus ranges from the sleep-quality review:

- 0 to 30 minutes: 100.
- 45 minutes: 75.
- 60 minutes: 40.
- 90 minutes or more: 0.

A very short latency should not be penalised here. It may reflect high sleep pressure, but duration and recent deficit already describe that situation more directly.

The second part of this component is circadian alignment. This should compare actual midsleep with the midpoint of the target sleep window using circular time:

- Within 30 minutes: 100.
- 60 minutes: 90.
- 90 minutes: 75.
- 120 minutes: 50.
- 180 minutes or more: 0.

When both values are available they can be averaged equally. If the device only supplies detected sleep start, latency remains missing and alignment carries the component. There should be no universal penalty simply for sleeping between 2am and 5am.

That may be badly aligned for one person and completely normal for a night worker. Travel and timezone shifts should reduce confidence until a stable local pattern returns.

Calculating overnight physiology

Overnight physiology should behave as supporting evidence. HRV, RHR, respiratory rate, oxygen saturation and temperature are valuable for understanding the night, but they do not directly measure whether the person was asleep or whether that sleep was continuous.

Reviews also find that objective sleep measurements explain only part of how people rate their own sleep, with results varying substantially between studies. [18]

The component should begin at 100 and only move down when comparable overnight signals leave the personal ranges defined in the accompanying Personal Metric Baselines paper. The signals can be grouped so correlated measurements do not each pretend to be independent:

- Autonomic group: HRV and resting heart rate.
- Respiratory group: respiratory rate and oxygen saturation.
- Temperature group: overnight temperature variation.

One abnormal group gives a physiology score of 80, 2 groups give 60, and all 3 give 40. A group becomes abnormal when one signal crosses its metric-specific alert boundary on 2 consecutive nights, or when 2 related signals cross together on the current night.

A severe absolute oxygen-saturation reading or a provider quality warning can create an immediate respiratory flag, although this remains a wellness warning rather than a diagnosis. Missing physiology stays missing and creates no score penalty. It lowers confidence instead. The physiology component can therefore reduce the final score by at most 3 points in normal use and cannot rescue poor sleep.

Calculating sleep stages

Sleep stages account for 5% of the score. The companion paper defines the validity checks: at least 90% timeline coverage, agreement with total sleep, valid non-overlapping intervals, a stable provider and algorithm source, and enough nights for a personal reference. These checks should be applied before the stage component is calculated. Once the data passes them, the stage score can be derived from personal deviation and broad age-based plausibility limits.

Population stage values are best used as plausibility rails. They change with age and they come from polysomnography rather than a wrist classifier. Across adulthood, N1 (light sleep) and N2 (stable light sleep) generally rise while slow-wave and REM sleep fall, and healthy adult sleep efficiency also declines with age. [16] [19]

The sleep-quality consensus found REM around 21% to 30% and N3 (deep sleep) around 16% to 20% compatible with good sleep in adults aged 26 to 64. It found N3 around 20% to 25% compatible with good sleep in teenagers.

It also found that REM at 41% or more was not a good-quality marker in young adults, adults or older adults, and N3 of 5% or less was not a good marker from the teenage years through adulthood. The panel did not agree on many positive stage ranges, particularly in older adults, which is a useful warning against building a detailed age quota table from weak evidence. [14]

The stage component should therefore compare REM and N3 percentages with the provider-specific 28-night personal reference first. We can use the centre and spread from the baseline paper:

- Both stages within 2 spreads of their references: 100.
- One stage between 2 and 3 spreads away: 80.
- One stage beyond 3 spreads, or both stages beyond 2 spreads: 60.
- The same stage remains beyond 2 spreads for 3 consecutive valid nights: 40.

Age plausibility acts as a cap. For ages 13 to 17, REM at 10% or less or N3 at 5% or less caps the stage component at 60. For ages 18 to 64, REM at 41% or more or N3 at 5% or less also caps it at 60.

For ages 65 and older, REM at 41% or more caps it at 60, while there is no hard low-N3 cap because the consensus evidence was not strong enough to set one. An impossible or internally inconsistent stage result should have failed the earlier quality checks and remain missing.

These are broad guardrails, not targets to chase. A teenager does not fail sleep just because N3 was 18% instead of 20%, and an adult does not receive bonus points for forcing REM into a narrow band.

Valid stages can support the picture or reduce the component when they are persistently strange. Missing stages use a neutral value of 100 in the score calculation and reduce stage confidence; they never become zero minutes and never transfer their full 5% to a less reliable input.

Combining the components

The principles above can be written as the following calculation:

$$\text{core score} = \text{weighted mean}(\text{duration } 35, \text{ regularity } 25, \text{ continuity } 20, \text{ timing/onset } 10)$$

$$\text{sleep score} = 0.90 \text{ core score} + 0.05 \text{ physiology score} + 0.05 \text{ stage score}$$

If physiology or stages are missing, their value in this formula is 100 so missing optional data cannot lower the sleep score. The confidence label records that they were absent. Duration is required.

At least 2 of regularity, continuity and timing/onset must also be available; otherwise a complete sleep score should be withheld. If one core component is unavailable, the available core weights are rescaled inside the 90-point core only.

Some nights also need caps. A duration shortfall of 3 hours caps the final sleep score at 60, 4 hours caps it at 40, and 5 hours or more caps it at 25. WASO of 90 minutes or more, maintenance efficiency below 65%, or 4 or more long awakenings caps the final score at 60. These caps preserve the principle established above: an attractive REM estimate cannot rescue a very short or severely fragmented night.

Missing data and confidence

Confidence should sit beside the result and never enter the scoring arithmetic. High confidence requires a valid main sleep, at least 90% sleep-period coverage, all 4 core components, a stable provider source and a 28-night personal reference. Moderate confidence has the essential sleep

record and at least 3 core components, but only 7 to 27 comparable nights or missing optional data.

Low confidence means the score had to use sparse history, a recent device change, incomplete sleep intervals or only the minimum 3 core components. Missing duration, an invalid main-sleep session or fewer than 2 supporting core components means no score.

Conclusion

Taken together, the sleep score supported by this research has the following form:

- Duration accounts for 35% and is scored from the shortfall against core sleep need after the 30-minute allowance. A 2-hour shortfall scores 65, 3 hours scores 35 and 5 hours scores 0.
- Regularity accounts for 25% and uses circular bedtime and wake-time drift from the correct personal day-type reference. Drift up to 30 minutes scores 100, 90 minutes scores 75 and 3 hours scores 20.
- Continuity accounts for 20% and is driven by WASO. Up to 20 minutes scores 100, 40 minutes scores 75 and 90 minutes scores 0, with maintenance efficiency and long awakenings used as caps.
- Timing and sleep onset account for 10%. Latency up to 30 minutes scores 100, 60 minutes scores 40 and 90 minutes scores 0. Midsleep alignment uses a separate personal target window and carries no universal late-night penalty.
- Overnight physiology accounts for 5% and works as a grouped personal-baseline penalty. It cannot improve the score and missing physiology causes no penalty.
- Sleep stages account for 5% and are only used after the quality rules in the companion paper pass. They use a provider-specific 28-night reference with broad age plausibility caps, never REM or deep-sleep debt.
- Duration is mandatory, optional inputs remain neutral when missing, severe duration and fragmentation apply explicit caps, and confidence is reported beside the score rather than becoming part of it.

This produces a sleep score in which every component has a defined role and no less reliable measurement can overwhelm the core sleep features. The exact interpolation curves and caps are design values built around the strongest thresholds available, so they should be validated against morning sleep quality and daytime function.

The remaining uncertainty is explicit. Some values come directly from consensus research, others depend on personal baselines, and the interpolation curves and caps are initial implementation choices.

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